

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

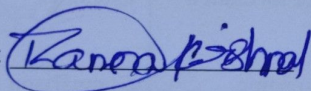
Total Marks: 25

Code of Conduct

I 192372280 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

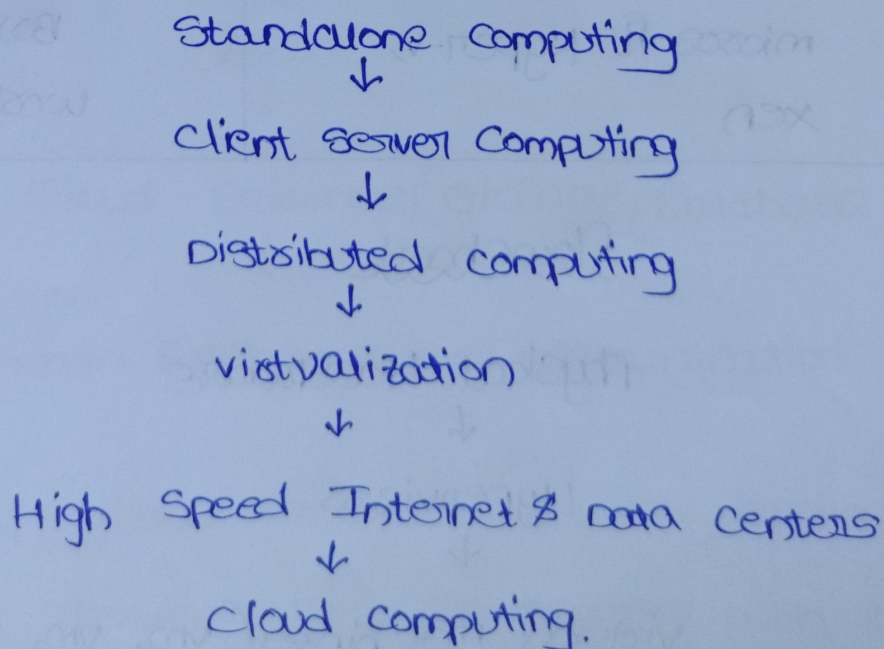
Assignment - 1

1. Describe the evolution from standalone computing to cloud based systems.

Ans:-

- * Standalone computer stores data locally
- * Client-server shares resources through a server
- * Distributed computing connects multiple computers
- * virtualization creates multiple virtual machines
- * Internet and data centers enables remote access
- * cloud computing provides on-demand services over the internet.

Flow chart:-

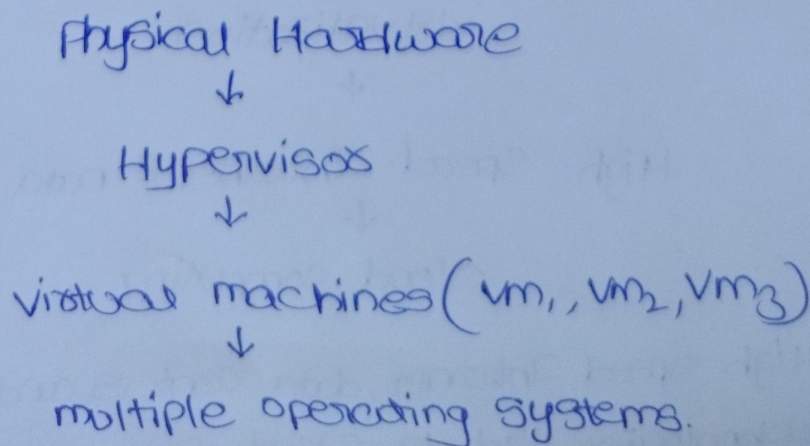


- * High speed Internet, data centers and virtualization technologies led to cloud computing.

2. Interpret the working of hypervisors in virtual difference b/w Type-1 and Type-2 Hypervisors examples.

Type-1 Hypervisor	Type-2 Hypervisor
* Runs directly on hard ware * Faster and more secure * Used in data centers * Ex:- VMware, ESXi, microsoft Hyper-V, xen	* Runs on top of an operating system * Easier to install but slower * Used for personal computers & testing * Ex:- oracle, virtual Box, VMware workstation.

Flowchart

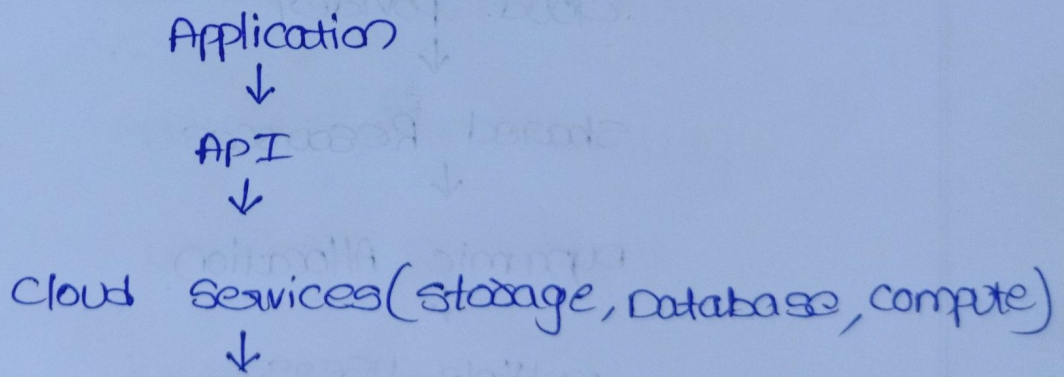


Role of APIs in cloud Service Integration

Benefits of API:-

- * connect different cloud services
- * Enable data sharing between applications
- * automate cloud operations
- * Improve Interoperability between different platforms
- * Support integration of third-party services.

Flow chart:-



Data Exchange & Communication

points:-

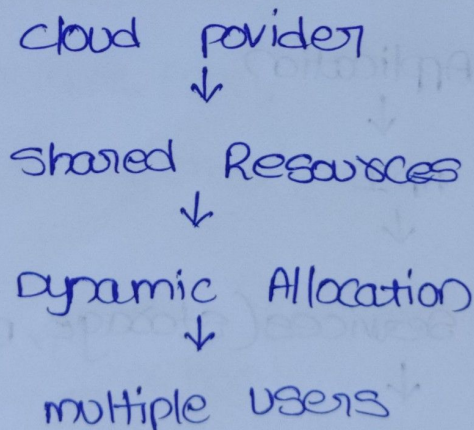
- * API's allow different software applications and cloud services to communicate with each other
- * They Enable developers to access cloud resources such as storage, database, and computing services.

4. Explain the concept of resource pooling in Computing. Illustrate how it benefits multiple users.

Benefits:-

- * Efficient utilization of resources
- * Reduces operational costs
- * Supports multiple users simultaneously
- * Provides better scalability and availability
- * Users receive resources without knowing their exact physical location.

Flow chart:-



points:-

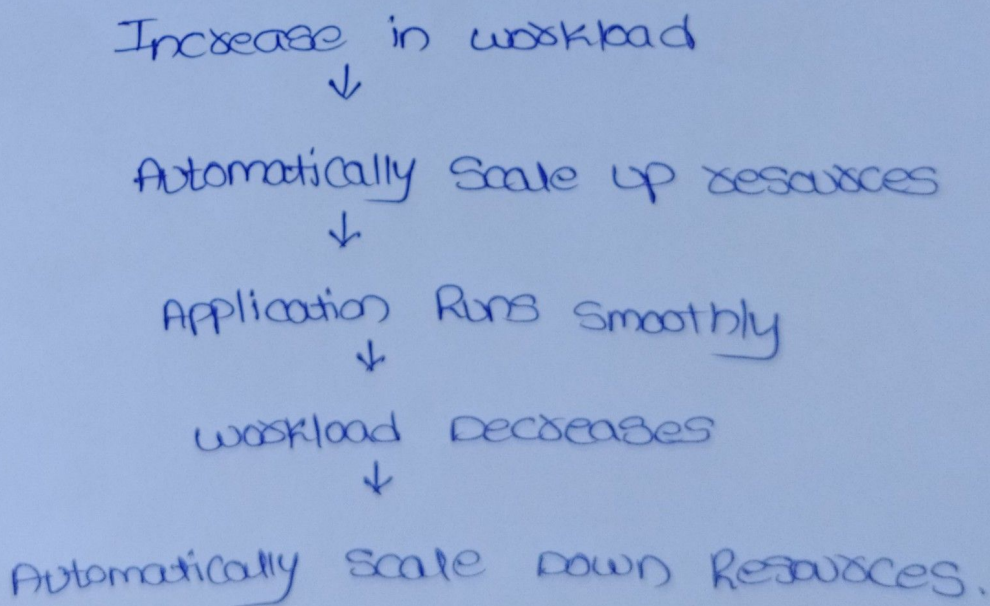
- * Resources are shared among users
- * Allocated based on demand
- * Improves resources utilization
- * Reduces operational cost
- * Provides scalability & Availability.

Rapid Elasticity in cloud computing

Benefits:-

- * Handles sudden increases in traffic
- * Reduces costs by using only required resources
- * Improves application performance
- * Ensures high availability
- * Supports dynamic and changing workloads efficiently

Flow chart



points:-

- * Automatically Increases resources
- * Reduces resources when demand decreases
- * Handles dynamic workloads
- * Improves performance.